

JAWAHAR R

Aspiring Data Engineer · AI & Data Science Student

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SUMMARY

Data Engineering student who builds and ships production-style pipelines rather than just studying them — including FlowMatrix, a Kafka-to-dbt e-commerce pipeline benchmarked against Rakuten Ichiba's data model, deployed live with automated data-quality checks. Comfortable across ingestion (Kafka, REST APIs), transformation (PySpark, dbt), orchestration (Airflow), validation (Great Expectations), and serving (FastAPI, PostgreSQL, Streamlit). Certified across IBM, Google Cloud, Harvard (CS50), and NPTEL. Targeting a Data Engineering internship where pipeline reliability and clean data modeling matter as much as shipping fast.

SKILLS

- **Languages:** Python, SQL, Java (basics), JavaScript, HTML/CSS
- **Data Engineering:** ETL/ELT, Apache Kafka, PySpark, dbt, Apache Airflow, Great Expectations, Data Modeling, Real-Time Processing
- **Databases & Cloud:** PostgreSQL, Neon Serverless, Docker, Google Cloud Platform, Render.com
- **Frameworks:** FastAPI, Uvicorn, REST API Design, Streamlit **AI/ML:** NLP, Machine Learning, Generative AI

EXPERIENCE

Data Analytics Virtual Internship

AICTE + EduSkills, supported by Google Cloud | Jun – Aug 2026 (8 weeks)

- Completed an 8-week Data Analytics Virtual Internship under the National Internship Portal (AICTE + EduSkills), supported by Google Cloud.

AI-ML Virtual Internship

AICTE + EduSkills, supported by Google for Developers | Oct – Dec 2025 (10 weeks)

- Completed a 10-week AI-ML Virtual Internship; presented the Live City Pulse data pipeline at the national AI For Bharat Hackathon and college project expo.

PROJECTS

FlowMatrix — Real-Time E-Commerce Data Pipeline | Kafka, PySpark, dbt, Airflow, Great Expectations, PostgreSQL | flowmarics.streamlit.app

- Built an end-to-end ELT pipeline (Kafka → PySpark → dbt → PostgreSQL) modeled on the Olist dataset as a Rakuten Ichiba proxy; processed 1,247 streaming events into 7 dbt models with 34 automated data-quality tests (100% pass rate).
- Orchestrated ingestion-to-dashboard refresh via a 6-task Airflow DAG with automated retry logic, achieving end-to-end pipeline runtime under 2 minutes; deployed a live Streamlit front-end backed by PostgreSQL (Neon Serverless).

Real-Time 3D Asset Data Pipeline & Serving API | Python, FastAPI, Three.js, NLP

- Built an ETL pipeline with FastAPI: streamed binary .glb assets over HTTP, parsed prompts via NLP keyword-matching into metadata queries, and served them through a thread-safe cache, cutting repeat-load latency to near-zero.

Weather ETL Dashboard | JavaScript, OpenWeatherMap API • Kisan-Sahayak — AI Agricultural Assistant | Python, NLP, Translation API

- Client-side ETL dashboard streaming live weather/AQI data for 5 cities, zero-backend, deployed on GitHub Pages.
- AI assistant (AI For Bharat Hackathon, national-level) delivering real-time crop prices and farming advice in Tamil and Hindi via an NLP translation pipeline.

CERTIFICATIONS

- **CS50: Introduction to Databases with SQL** — Harvard University (edX) — 2026
- **Introduction to Data Engineering** — IBM / Coursera — May 2026
- **Python for Data Science, AI & Development** — IBM / Coursera — Jul 2026
- **Python Project for Data Engineering** — IBM / Coursera — Jul 2026
- **Google Cloud Data Analytics Certificate** — Google Cloud — Jul 2026
- **Computer Architecture & Organization** — NPTEL / IIT Kharagpur — Jul–Oct 2025

EDUCATION

B.Tech, Artificial Intelligence & Data Science | 2024 – 2028 | Dhanalakshmi Srinivasan Engineering College (Autonomous), Perambalur, Tamil Nadu

CGPA: 8.5 (through 4th semester)

LANGUAGES

Tamil (Native) • English (Advanced) • Japanese (Beginner — preparing for JLPT N5)